

Prajwal Mhatre (MS)

◆ prajwal7649@gmail.com ◆ [Linkedin](#) ◆ [GitHub](#)

EDUCATION

Clemson University, South Carolina, USA

Completed – Dec 2024

Masters in Electrical and Computer Engineering

CGPA – 3.7/4

- **Courses** : Embedded Systems , Parallel Computing and Architecture, AI in wireless communications , AI, Advanced Linear Algebra, Data Science, Data Analysis, Representation Learning.

Mumbai University, India

Completed – Aug 2022

Bachelors in Electronics and Telecommunication Engineering

CGPA – 7.52/10

- **Courses** : Database Management Systems, Big data Analysis, Micro-Controllers, Micro-Processors, Communication Engineering.

WORK HISTORY

OpenQQuantify

Software Developer Intern

Aug 2024 – Aug 2025

- Built a multi-agent LLM system (**Python, Flask**) where a **supervisor agent** routes requests to 7 specialists agents, turning one plain-English prompt into a full embedded build package: BOM, pin netlist, firmware, schematic, assembly guide.
- Benchmarked **vector RAG** (LlamaIndex, ChromaDB, 2 embedding models, chunk sizes 5,000–512) and replaced it with exact-match SQL retrieval injected as hard prompt constraints, eliminating pin-level hallucination.
- **Patched** the Schemdraw library across 4 files to emit addressable **SVG** element IDs, converting static generated schematics into interactive diagrams where clicking any component or wire isolates it.
- Modularized the frontend into 9 single-responsibility JS modules and built CI on **GitHub Actions** running 119 Python and JS tests, lint, and secret scanning on every push.
- Shipped to production as the company's flagship product, presented by the founder in the seed-round pitch deck.

Clemson University

Software Research

Jan 2024 – Aug 2024

- Built a custom **distributed neural network** using **Python, PyTorch** and **CUDA**, enabling power usage benchmarking against standard models on a single GPU.
- Designed an **optimizer-agnostic training layer** that treats each sub-network's optimizer as a pluggable component, currently supporting Particle Swarm Optimization and gradient descent.
- Engineered a GPU power-instrumentation pipeline that logs per-run energy consumption, enabling like-for-like **accuracy-vs-energy** comparison against monolithic time-series baselines (RNN, LSTM, dense MLP).
- **Created a visualization tool** to convert logged power consumption metrics into clear charts, grounding architecture decisions in measured data.

Clemson University

Teaching Assistant - Signals and Systems

Aug 2023 – May 2024

- Graded 100+ students, 3 assignments per student each week and 1 MATLAB assignment task every two weeks.

PROJECTS

IoT-Based Fire Hazard Mitigation Robot (Embedded C | ESP32-CAM | Arduino Uno | Android Studio | Firebase)

- Architected a **dual-microcontroller embedded system** utilizing an **Arduino Uno** for real-time sensor polling (DHT11, MQ-6) and relay actuation, interfaced via UART to an **ESP32-CAM** acting as a dedicated Wi-Fi and video streaming node.
- Developed a native companion application in **Android Studio (Java/XML)** to serve as a remote telemetry dashboard, rendering real-time, low-latency video streams and **pushing asynchronous motor control**.
- Engineered a cloud-integrated media pipeline using **Firebase** Realtime Database to persist device telemetry and integrated third-party libraries to record, upload, and playback operational live streams for post-event analysis.

Real-Time Payment Fraud Detection Engine (Java 21 | Spring Boot | React | SQL)

- Built an async fraud-screening pipeline (bounded queue + worker thread pool) with explicit back-pressure instead of blocking HTTP threads.
- Designed a weighted rule engine (amount, velocity, merchant risk) with graph traversal detecting circular fund flows between accounts.
- Secured REST endpoints with stateless JWT auth and role-based access; unit + integration tests across rules, engine, and queue.

SKILLS

- **Programming Languages**: Python, SQL, C#, Java, R, PowerShell, Shell, JS, Bash, MATLAB, Embedded C, ARM Assembly, win-Batch
- **Databases**: MySQL, MongoDB, NoSQL, ChromaDB
- **Tools** : Git, Linux, Android Studio

ACHIEVEMENTS

Awarded “Prime” Cadre via TCS National Qualifier Coding Test (NQT) | 2026